hovered about the 36,000 KBps mark, with the read and write speeds being more or less within the 52 - 54 MBps range. This clearly indicates that SATA needs to make a significant technology advance in order to beat the prevailing ATA standard. As of now, the ATA 133 standard still rocks.

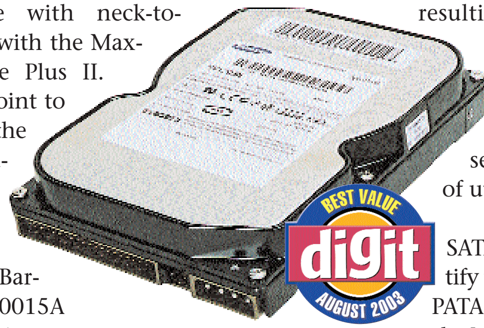
File transfer tests

This test was introduced in our hard drive comparison tests to see how they fared in a real world environment. Any user with access to a PC uses it to transfer data in some way or the other. The hard drive plays an important role in this, since it is the data storage component. The faster the rate of data transfer, the better the hard drive.

The test file comprised of two sets of files. An assorted set of files, comprising of mp3 files, software installation setups, documents, and zip files, all of which amounted to 1 GB, were put in a single folder. These were then transferred from the main hard drive to the test hard drive, and then from one partition to the other on the test hard drive. A 1 GB movie file was also taken for the tests. The assorted set of files, and the movie file simulated the random, and sequential read and write speeds for the hard drive. The time taken to transfer the files was noted down as the result.

The results for these tests were quite similar all over. In the SATA category, the Maxtor DiamondMax Plus 9 200 GB drive showed the best results in this test by completing the entire file transfer in just 57, and 52 seconds for random and sequential read and write respectively. This hard drive is really good for people who need to transfer a lot of data, onto or off the hard drive. The Seagate's ST3160023AS took the second spot, by taking a slightly longer time than the Maxtor's 200 GB drive, but, nevertheless, put up a respectable score.

In the ATA category, nothing could beat the Maxtor Maxline Plus II 250 GB drive in the real world tests. This drive lost out to its SATA sibling only because of its slightly higher CPU utilisation, taking some extra seconds to complete the copying of the file. This is a good point to show how even a slight difference in the CPU utilisation matters in the real world. The other drives, such as the ones from Samsung, and especially the SP0802N, displayed good real world performance with neck-to-neck scores with the Maxtor MaxLine Plus II. One good point to note is that the CPU utilisation of both the drives was on par. The Seagate Barracuda ST340015A was the biggest disappointment in the entire test. It was the only drive that came up with worst results.



Samsung SpinPoint V Series SV1203N

Photoshop test

In this test, we note down the time required to load Photoshop after its installation, and the time taken to open a 200 MB file. The drive with the higher buffer memory, and good read burst speed will definitely score better than other drives. A lot depends on the read ahead memory, and the drive controller, present on the drive itself. The Maxtor MaxLine Plus II 250 GB—which took just 4 seconds to load Photoshop, and needed just 29 seconds to load the 200 MB test file—walked away with the crown. This can be attributed to the 8 MB buffer memory, and a good read speed burst of 9 MBps. We recommend it to anyone who wants a good drive for video editing. The Samsung SV1203N ATA drive was second in line, taking just 5 seconds, and 28 seconds to open Photoshop and the test file. Other drives such as the Seagate Barracuda ST3160021A, and the Samsung SP0802N, took around the same time.

All these drives have excellent onboard drive controller electronics resulting in high performance. The test results seem to be very much alike, with hardly any variation in them. However, even a second's difference could be of utmost importance.

Unfortunately, none of the SATA drives gave results that justify them as the successors of the PATA technology. Coming in first, was the Maxtor 120 GB DiamondMax plus 9 drive, sporting a score that was neck-to-neck with the winner in this particular test. All other drives took longer times to load the same image file.

To sum it up...

The test concluded with some great insights, toppling some of the myths about the ATA 133 standards capability, and was an eye opener regarding facts about the SATA technology. Although many motherboard, and chipset manufacturers deny official support to the ATA 133 standard under the pretext of instability, the fact remains clear that even today, it has the capability to beat newer technologies such as SATA, which is in its

Decision Maker

	Entry Level Hard Disk	Good balance of performance and price	Super Fast and scalable Hard Disk
You Need	A steady going HDD that can store all your Movies, MP3's and other data with decent performance and a large enough capacity	A HDD with good performance, with large storage capacity, taking less time to copy files on and off the HDD. A Drive that is future proof and won't become obsolete in the wake of SATA becoming a standard.	A super fast drive with excellent performance, low latency, takes less time to copy data on and off the HDD and also supports future satandards.
Look For	A 7,200 RPM drive with atleast 40 to 80 GB of capacity with price per MB in the region of 5 to 7 paisa	A 7,200 RPM drive with 120 GB to 160 GB of space and providing 8 MB of buffer and having excellent read write speeds.	HDD with atleast 8 MB of buffer, spindle speed over 7,200 rpm, having extremely low CPU utilisation and having good over all performance with loads of real estate to store you data.
Our Pick	The Samsung SV0802N, SV0602H, SP0411N	The Maxtor DiamondMax plus 9 120, 160 GB and Samsung's SV1203N	The Maxtor Maxline Plus II, Maxtor DiamondMax Plus 9 200 GB
Price	Rs 4,500 to Rs 6,200	Rs 7,500 to Rs 12,000	Rs 15,000 and above